Monotonicity of the signature of a mixed graph matrix

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MathDB 364187. Task 1: a full proof. We prove Conjecture 1 of Bonetto and Jardón Kojakhmetov [1, p. 7], also retained as Conjecture 3.1.1 in [2, p. 55]. The proof applies to the more general symmetric matrix family proposed immediately after their conjecture.

For a real symmetric matrix Q, let $n_+(Q)$, $n_0(Q)$ and $n_-(Q)$ be its positive, zero and negative eigenvalue counts, with multiplicity. Its signature is $s(Q) = n_+(Q) - n_-(Q)$. Write $X \preceq Y$ if $Y - X$ is positive semidefinite.

Theorem 1. *Let A, B be real symmetric $n \times n$ matrices, with $A \succeq 0$. For $M(t) = A + tB$ and $0 < u < v$,*

$$n_+(M(v)) \le n_+(M(u)), \qquad n_-(M(v)) \ge n_-(M(u)). \tag{1}$$

Consequently $s(M(v)) \le s(M(u))$, including at singular parameter values.

Lemma 2. *If X, Y are real symmetric matrices and $X \preceq Y$, then $n_+(X) \le n_+(Y)$ and $n_-(X) \ge n_-(Y)$.*

Proof. The number $n_+(Q)$ is the largest dimension of a linear subspace on which $x^T Q x$ is strictly positive for every nonzero x. Indeed, the positive eigenspace has that dimension. Any subspace of larger dimension intersects the span of the nonpositive eigenvectors nontrivially, so cannot be positive definite. The analogous statement holds for $n_-(Q)$.

If $x^T X x > 0$ on a subspace away from zero, then $x^T Y x \ge x^T X x > 0$ there. This proves the assertion for n_+. If $x^T Y x < 0$ on a subspace away from zero, then $x^T X x \le x^T Y x < 0$ there, proving the assertion for n_-. Zero eigenvalues cause no change to either argument. □

Proof of Theorem 1. Positive scalar multiplication preserves all three inertia counts. Normalize the matrix by its parameter:

$$N(t) = \frac{M(t)}{t} = B + \frac{A}{t}.$$

For $0 < u < v$ we have

$$N(u) - N(v) = \left(\frac{1}{u} - \frac{1}{v}\right) A \succeq 0. \tag{2}$$

Thus $N(v) \preceq N(u)$. Lemma 2 gives (1), since $N(t)$ and $M(t)$ have the same inertia. Subtracting the two count inequalities proves the signature inequality. □

Application to the source conjecture

In Definition 3 of [1], a simple undirected graph F has its edges partitioned between spanning subgraphs G and H, and

$$M(\epsilon) = L_G + \epsilon A_H, \qquad \epsilon > 0.$$

Here A_H is real symmetric, and the graph Laplacian satisfies

$$x^T L_G x = \sum_{\{i,j\}\in E_G} (x_i - x_j)^2 \geq 0.$$

Theorem 1 with $A = L_G$ and $B = A_H$ therefore proves the conjecture for every permitted graph and every positive parameter. In the associated gradient system $\dot{q} = -M(\epsilon)q$, the unstable eigenspace has dimension $n_-(M(\epsilon))$, which is nondecreasing. The stable dimension $n_+(M(\epsilon))$ is nonincreasing.

As in the source's step-function plots, decreasing means nonincreasing; constant intervals are allowed. The theorem includes the exact values at zero crossings and allows a common kernel for every parameter. It does not assert that each unnormalized eigenvalue is a decreasing real-valued function. The positive-parameter domain is precisely that of the source.

Verification and attribution

Lemma 2 is the standard order property of inertia, with an elementary proof included. The only further step is the normalization (2). The exact verifier checks all graph partitions through four vertices, general symmetric pencils, and singular and repeated crossings with a permanent kernel. Independent characteristic-root isolation checks the inertia routine. These finite tests supplement the unrestricted proof. Review is by the authoring agent only.

Proposed follow-up Tasks 2–5

Task 2. Determine the positive crossing parameters and their multiplicities from a canonical decomposition of the pencil (A, B), allowing a nontrivial common kernel.

Task 3. Characterize graph partitions (G, H) attaining each possible total increase of $n_-(L_G + \epsilon A_H)$ on $(0, \infty)$.

Task 4. Extend the inertia monotonicity statement to families $a(t)A + b(t)B$ with $b(t) > 0$, and characterize the coefficient functions that guarantee it for every $A \succeq 0$ and symmetric B.

Task 5. Determine hypotheses for an analogous monotonicity theorem for self-adjoint operator pencils, including domains and finiteness of negative spectral subspaces.

References

[1] R. Bonetto and H. Jardón Kojakhmetov, *On the Eigenvalues of Graphs with Mixed Algebraic Structure*, arXiv:2408.00487v1, 2024, Definition 3, Definition 8 and Conjecture 1.

[2] R. Bonetto, *Oscillations and diffusion in dynamic networks models*, University of Groningen PhD thesis, 2025, doi:10.33612/diss.1329052104, Section 3.1, pp. 47–58, especially Conjecture 3.1.1 on p. 55.